# MODULES ARE SMALL WITH RESPECT TO INJECTIONS FOR ORDINALS OF LARGE COFINALITY

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument; the opening remark that the proof is straightforward is followed by the argument itself, including the special case and the cofinality bound, so no step is left to the reader.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- Smallness with respect to injections. Same chapter, the definition of an object being alpha-small with respect to a class of morphisms. Supplies the property the proposition establishes, namely that every map from the module into a colimit of an inductive system factors through one stage.
- The finite case. Same chapter, Lemma lemma-out-of-finite. Supplies the special case of a finite module, which is treated first and which the general argument imitates.
- Cofinality and upper bounds in an ordinal. Set theory, the elementary facts about cofinality of an ordinal and upper bounds of small subsets. Supplies that a subset of an ordinal of cardinality at most kappa has an upper bound strictly below an ordinal of cofinality bigger than kappa.

*Source reference.* Original lemma 05NR: sources/stacks-injectives/pinned-original-source.tex, source lines 326–326. Navigation pointer invoked at proof line 400; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original equation 05NN: sources/stacks-injectives/pinned-original-source.tex, source lines 279–279. Navigation pointer invoked at proof line 327; the

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counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Proposition 0.1.** *Let $R$ be a ring. Let $M$ be an $R$-module. Let $\kappa$ the cardinality of the set of submodules of $M$. If $\alpha$ is an ordinal whose cofinality is bigger than $\kappa$, then $M$ is $\alpha$-small with respect to injections.*

*Proof.* The proof is straightforward, but let us first think about a special case. If $M$ is finite, then the claim is that for any inductive system $\{B_\beta\}$ with injections between them, parametrized by a limit ordinal, any map $M \to \operatorname{colim} B_\beta$ factors through one of the $B_\beta$. And this we proved in Lemma 05NR.

Now we start the proof in the general case. We need only show that the map (05NN) is a surjection. Let $f : M \to \operatorname{colim} B_\beta$ be a map. Consider the subobjects $\{f^{-1}(B_\beta)\}$ of $M$, where $B_\beta$ is considered as a subobject of the colimit $B = \bigcup_\beta B_\beta$. If one of these, say $f^{-1}(B_\beta)$, fills $M$, then the map factors through $B_\beta$.

So suppose to the contrary that all of the $f^{-1}(B_\beta)$ were proper subobjects of $M$. However, we know that

$$\bigcup f^{-1}(B_\beta) = f^{-1}\left(\bigcup B_\beta\right) = M.$$

Now there are at most $\kappa$ different subobjects of $M$ that occur among the $f^{-1}(B_\alpha)$, by hypothesis. Thus we can find a subset $S \subset \alpha$ of cardinality at most $\kappa$ such that as $\beta'$ ranges over $S$, the $f^{-1}(B_{\beta'})$ range over *all* the $f^{-1}(B_\alpha)$.

However, $S$ has an upper bound $\widetilde{\alpha} < \alpha$ as $\alpha$ has cofinality bigger than $\kappa$. In particular, all the $f^{-1}(B_{\beta'})$, $\beta' \in S$ are contained in $f^{-1}(B_{\widetilde{\alpha}})$. It follows that $f^{-1}(B_{\widetilde{\alpha}}) = M$. In particular, the map $f$ factors through $B_{\widetilde{\alpha}}$. □